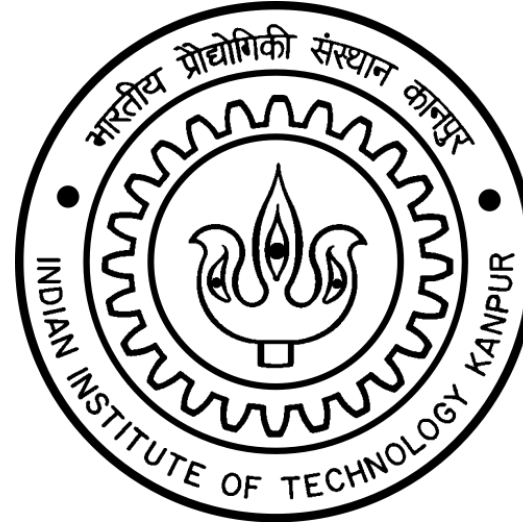




Synthesis and Characterization of Layered Ni-Nb-Te Single Crystals

PHY315: Modern Physics Laboratory - Course Project Presentation

Presentation by
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Introduction & Motivation

- Transition metal chalcogenides (TMCs) have been a cornerstone of condensed matter physics research for decades.
- Their often quasi-two-dimensional, layered structures host a rich variety of correlated electronic phases, including superconductivity [1], complex magnetism, and Charge Density Waves (CDWs) [2].
- The ability to substitute different transition metals (like Ni and Nb) into a telluride lattice offers a powerful route to tune these electronic ground states.

Sources:

[1] Yang et. al. Pressure-induced nontrivial Z₂ band topology and superconductivity in transition metal chalcogenide Ta₂Ni₃Te₅. *Phys. Rev. B* 107, L020503.

<https://doi.org/10.1103/PhysRevB.107.L020503>

[2] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. "Charge-density waves and superlattices in the metallic layered transition metal dichalcogenides". *Phys. Rev. Lett.* 32, 882. <https://doi.org/10.1103/PhysRevLett.32.882>

Project Goals

1. Goal 1: Synthesis

- a. To synthesize novel ternary Ni-Nb-Te single crystals using the Chemical Vapor Transport (CVT) method.
- b. The synthesis was initiated with a 1:2:2 (Ni:Nb:Te) stoichiometric ratio

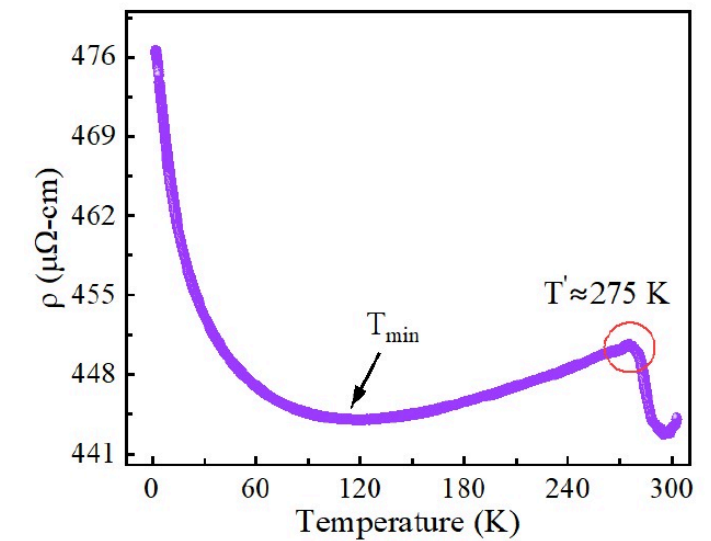
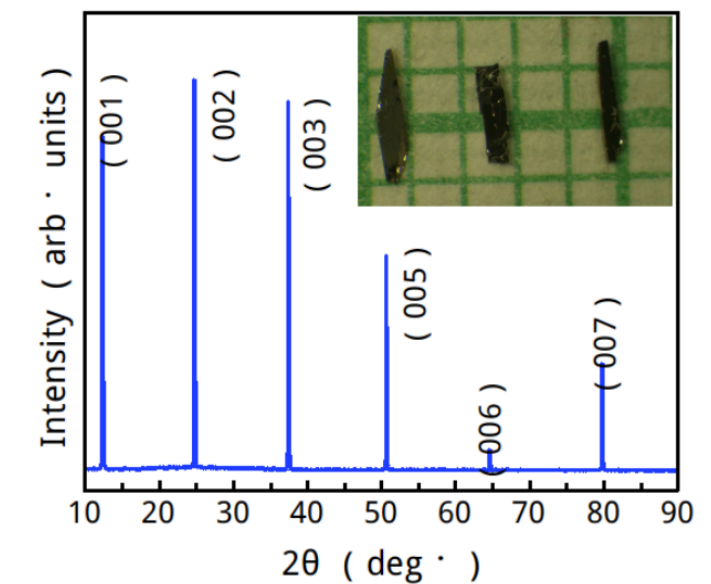
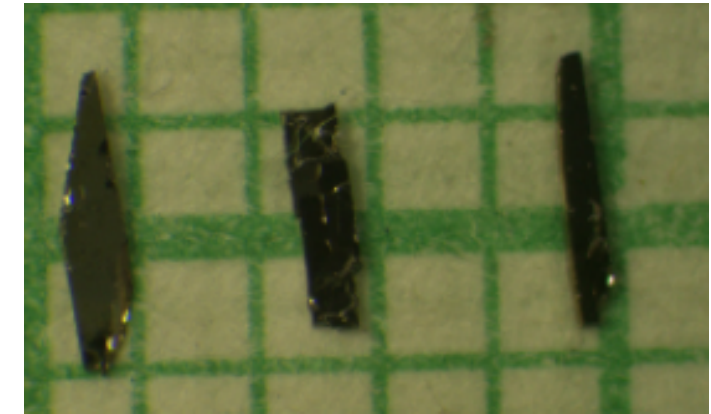
2. Goal 2: Characterization

- a. To determine the true elemental composition and crystalline structure of the resulting crystals using Energy Dispersive X-Ray Spectroscopy (EDX) and X-Ray Diffraction (XRD).
- b. To probe the electronic properties of the material via temperature-dependent resistivity ($R(T)$) measurements.

3. Goal 3: To identify and analyze any observed electronic or structural phase transitions and quantum transport phenomena based on the characterization data.

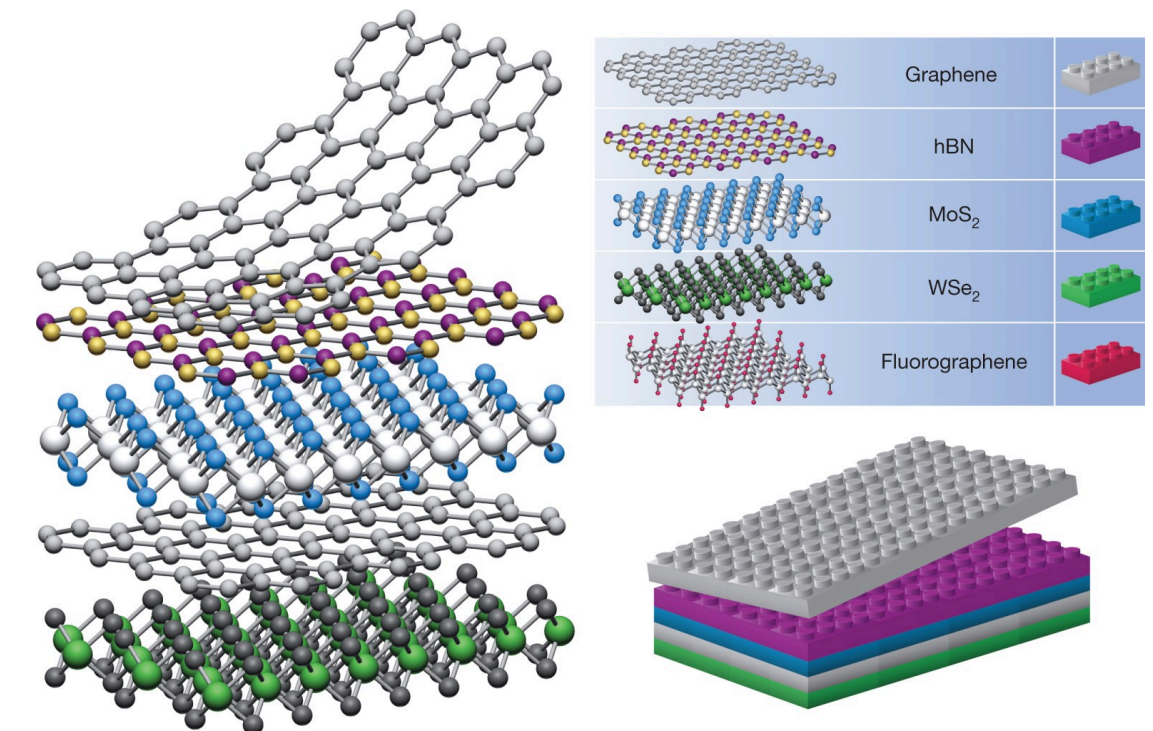
Summary of Results

1. **Synthesis:** Successfully synthesized high-quality, cleavable single crystals
2. **Composition (EDX):** The grown crystals are non-stoichiometric, with an average composition of Ni: 1 , Nb: 1.05 , Te: 2.45 . This is a Te-rich phase of the 1:1:2 (NbNiTe_2) structure.
3. **Structure (XRD):** The XRD pattern exhibits only sharp (00l) diffraction peaks, confirming a highly-oriented, single-crystalline, c-axis oriented crystals.
4. **Electronic Transport ($R(T)$):** The material is metallic and displays two significant anomalies:
 - a. A sharp resistivity drop at $T' \sim 275 \text{ K}$
 - b. A resistivity minimum at $T_{\min} \sim 111.5 \text{ K}$



Background: Van Der Waals Materials

- Layered solids composed of 2D atomic sheets; easily cleavable
- Anisotropic Bonding:
 - Intra-layer: Strong covalent or ionic bonds (e.g., the Nb-Te bonds).
 - Inter-layer: Weak van der Waals forces (attraction between induced dipoles).
- Conductivity within the plane is much higher than between planes.
- Possible to observe:
 - Charge Density Waves
 - Superconductivity
 - Magnetism
 - Klein Tunneling
 - A half integer shift in the quantum Hall effect



Source: Geim, A., Grigorieva, I. Van der Waals heterostructures. *Nature* 499, 419–425 (2013).
<https://doi.org/10.1038/nature12385>

Literature Review: Similar Materials

1. **NbNiTe₂** (Base Compound):

- Exhibits a structural phase transition (Orthorhombic → Monoclinic) at $T = 373 \text{ K}$ ($100 \text{ }^{\circ}\text{C}$).
- The transition is driven by steric effects.
- Literature reports no magnetic ordering.

2. **NbCoTe₂**: Reported to be a metal and Pauli-paramagnetic (non-magnetic, with no local moments).

3. **NbFeTe₂**: Exhibits antiferromagnetic (AFM) ordering at $T_N \approx 20 \text{ K}$.

Insights:

- All of the above are layered vdW materials.
- The observed transition at 275 K does not match the known structural transition of NbNiTe₂ (373 K). This new phase is likely induced by the significant Te-excess and non-stoichiometry.

Neu et. al. Orthorhombic to monoclinic phase transition in NbNiTe₂. <https://link.aps.org/doi/10.1103/PhysRevB.100.144102>

Li et. al. New layered ternary niobium tellurides: Synthesis, structure, and properties of niobium metal telluride, NbMTe₂ (M = iron, cobalt) <https://pubs.acs.org/doi/10.1021/ic00032a024>

Background: Chemical Vapor Transport (CVT)

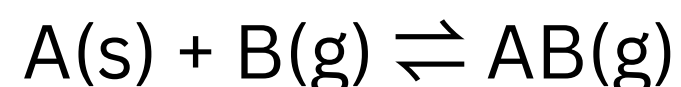
Problem: Material to be grown has high melting point and negligible vapor pressure, so it cannot be evaporated.

Solution: A Transport Agent (e.g., I₂, Cl₂)

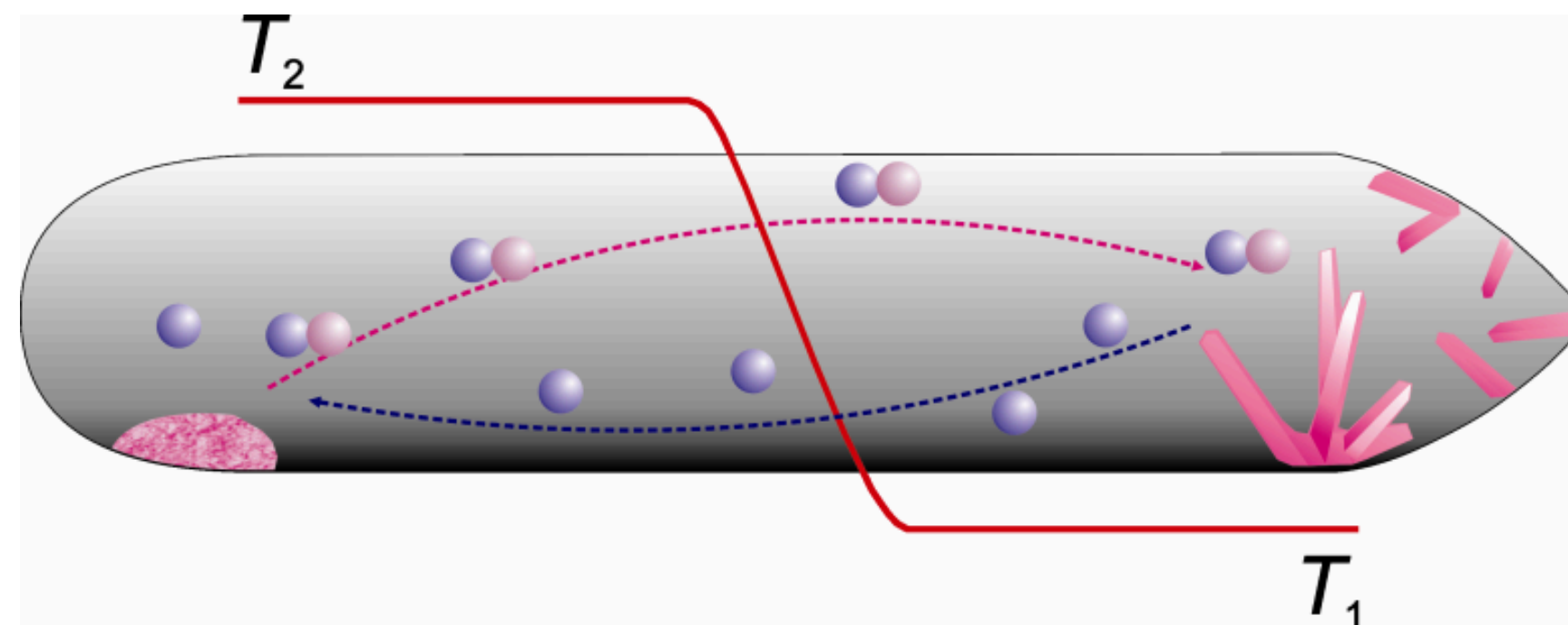
Source Zone (T₂): Volatile transport agent reacts with polycrystalline powder to form volatile gas complex

Transport: This gas diffuses along a sealed tube due to the concentration gradient

Growth Zone (T₁): At a T₁, the reaction reverses. The gas decomposes, depositing the pure crystal and releasing the transport agent, which then diffuses back to the source zone to repeat the cycle.



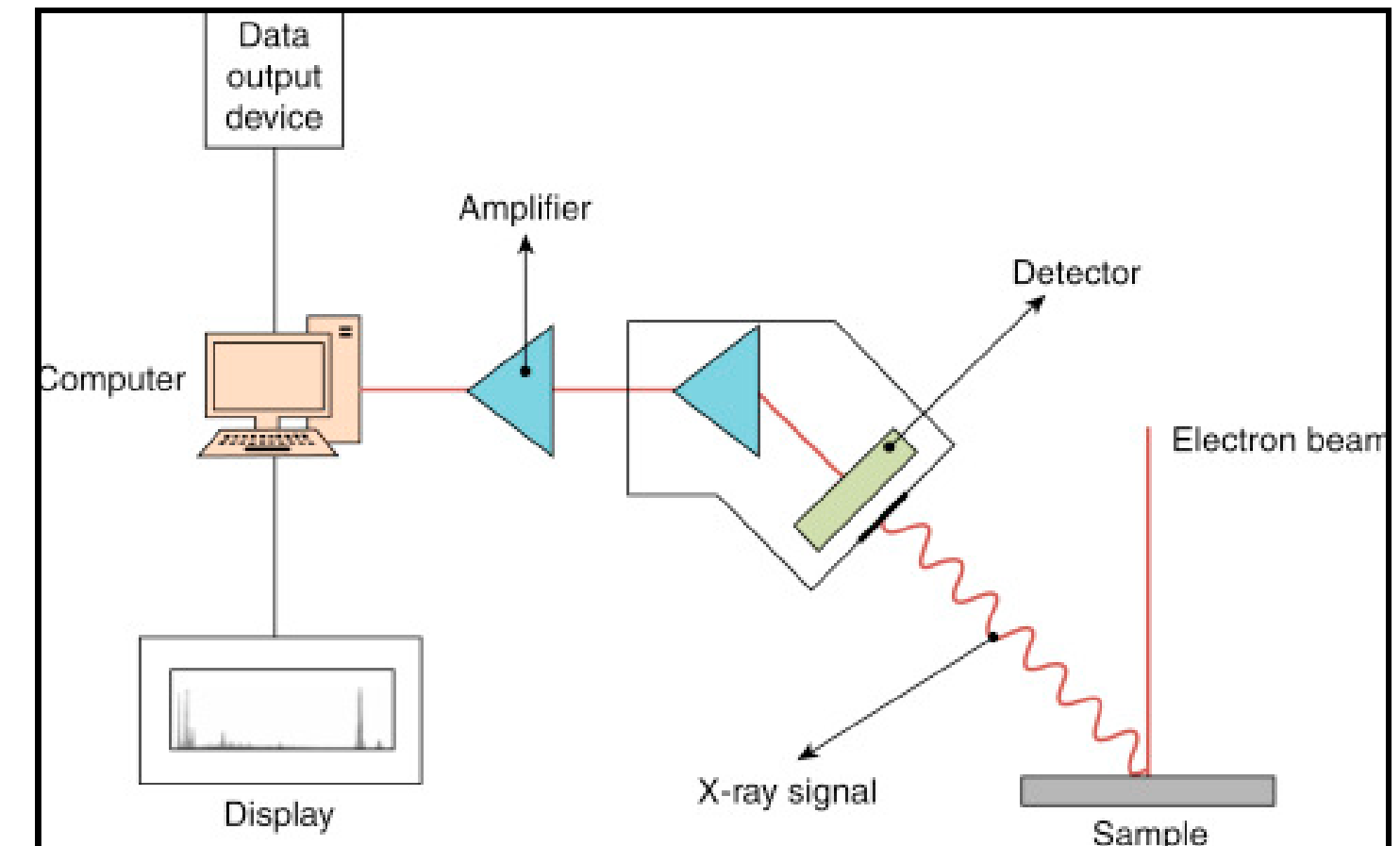
Source: Schmidt, Binnewies, Glaum and Schmidt
Chemical Vapor Transport Reactions-Methods,
Materials, Modeling. DOI: [10.5772/55547](https://doi.org/10.5772/55547)



Background: Energy Dispersive X-Ray Spectroscopy (EDX)

Purpose: To determine the elemental composition (stoichiometry) of a sample.

1. **Excitation:** A high-energy electron beam (in a Scanning Electron Microscope) hits the sample.
2. **Ionization:** The beam knocks out a core-shell electron
3. **Relaxation:** An outer-shell electron falls to fill the hole.
4. **Emission:** The energy difference between shells released as X-ray.
5. **Energy:** The energy of this X-ray is unique and characteristic of the element (Moseley's Law).
6. **Intensity:** The number of X-rays (counts) at that specific energy is proportional to the concentration (atomic %) of that element.



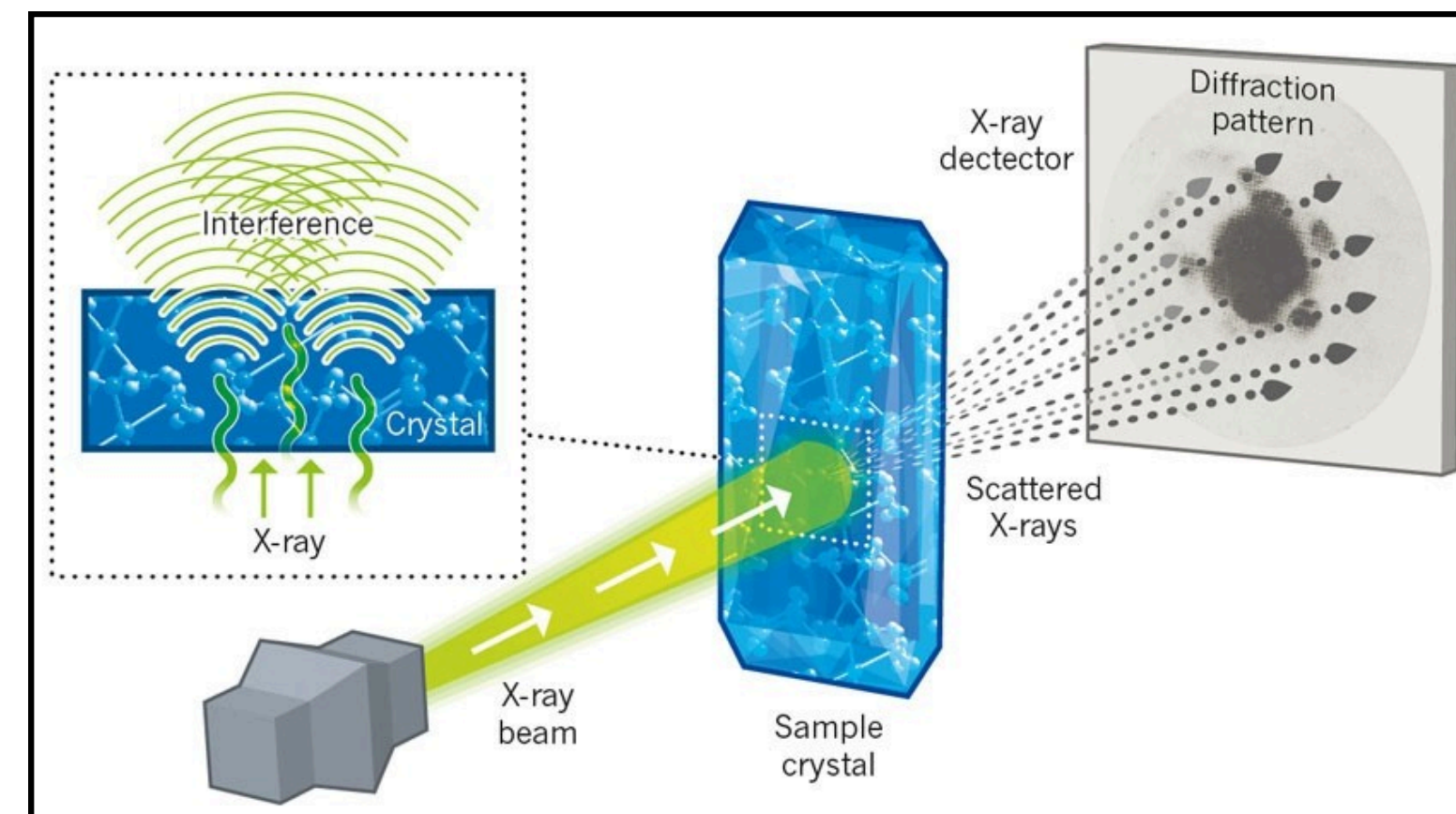
Source: Colpan, Nalbant and Ercelik. *Fundamentals of Fuel Cell Technologies*. <https://doi.org/10.1016/B978-0-12-809597-3.00446-6>

Background: X-Ray Diffraction (XRD)

Purpose: To determine the crystal structure of a sample.

1. X-rays are directed at a crystal.
2. The periodic atomic planes cause constructive interference at specific angles according to Bragg's law
3. **Peak Position (2θ)**: Tells the distance between atomic planes
4. **Peak Intensity**: Tells the arrangement and type of atoms within planes.
5. **Peak Pattern**: The unique set of all (2θ) peaks is a fingerprint of the crystal's specific phase, symmetry, and lattice parameters.

$$n\lambda = 2d\sin\theta$$



Source: Nicola Jones. Crystallography: Atomic secrets.
<https://doi.org/10.1038/505602a>

Background: Resistivity v/s Temperature Analysis

1. For a metal, ρ increases with T

Why: At $T > 0$ K, atoms vibrate (phonons). Electrons scatter off these vibrations, which causes resistance. More T = More vibrations = More scattering = Higher ρ

2. For a Semiconductor / Insulator, ρ decreases as T increases.

Why: Resistance is dominated by the number of charge carriers. At $T=0$ K, all electrons are locked in bands. As T increases, electrons are thermally kicked into the conduction band, creating carriers. More T = More carriers = Lower ρ

3. Anomalous Behavior (Quantum Materials):

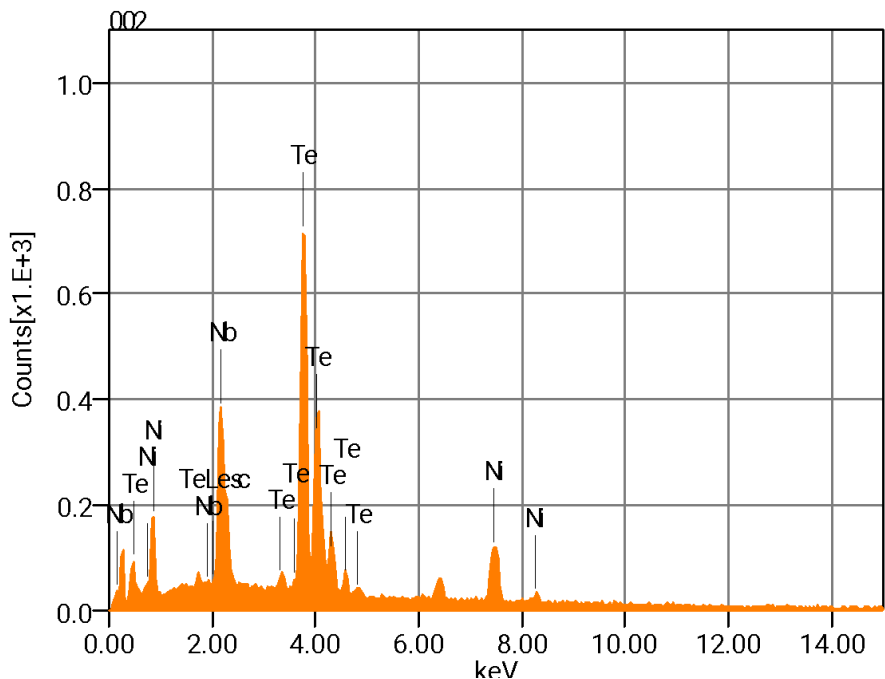
- Phase Transitions (e.g., CDW): Can cause a sudden hump or drop in resistivity as a new gap opens in the electronic structure.
- Low-T Upturns (e.g., Kondo): A minimum in ρ at low T, where resistivity increases as $T \rightarrow 0$ K. This is a signature of quantum scattering effects.

Crystal Synthesis

- **Method:** Chemical Vapor Transport (CVT)
- **Precursors:** High-purity (99.99%+) powders of Nickel, Niobium, and Tellurium.
- **Starting Stoichiometry:** 1 : 2 : 2 (Ni:Nb:Te)
- Mixture is sealed in quartz tube and heated to 900°C for 3 days to prepare polycrystalline powder samples
- **Transport Agent:** Iodine (I_2 , 5mg/cc)
- **The Apparatus:** A three-zone horizontal tube furnace.
- **The Ampoule:** High-purity quartz tube, cleaned and baked under vacuum.
- **Temperature Profile:**
 - Source Zone (T_2): 950°C
 - Growth Zone (T_1): 850°C
- **Duration:** 14 days
- **Results:**
 - Shiny, metallic, plate-like single crystals were grown in the Growth Zone.
 - Crystals are mechanically soft and easily cleaved with tape, confirming a vdW layered structure.



Characterization I: EDX (Composition)



Technique: Energy Dispersive X-Ray Spectroscopy (EDX) in an SEM.

Observation: Clear characteristic X-ray peaks for Nickel (Ni), Niobium (Nb), and Tellurium (Te)

Conclusion:

- Compound is not the 1:2:2 (NiNb₂Te₂) starting ratio.
- Compound is based on the 1:1:2 (NbNiTe₂) structure, but is highly non-stoichiometric.
- The two key features are:
 - Cation Ratio: The Ni:Nb ratio is ~1:1.
 - Tellurium Excess: The crystal is significantly Te-rich

Table 1: EDX compositional data of Ni, Nb, and Te for four points (P1–P4) and their averages.

Element	Area	P1	P2	P3	P4	Average
Ni	24.58	24.02	22.83	23.10	22.50	23.406
Nb	21.63	21.51	22.71	22.31	22.75	22.182
Te	53.79	54.47	54.46	54.59	54.75	54.412
Total	100	100	100	100	100	100

Characterization II: XRD (Structure)

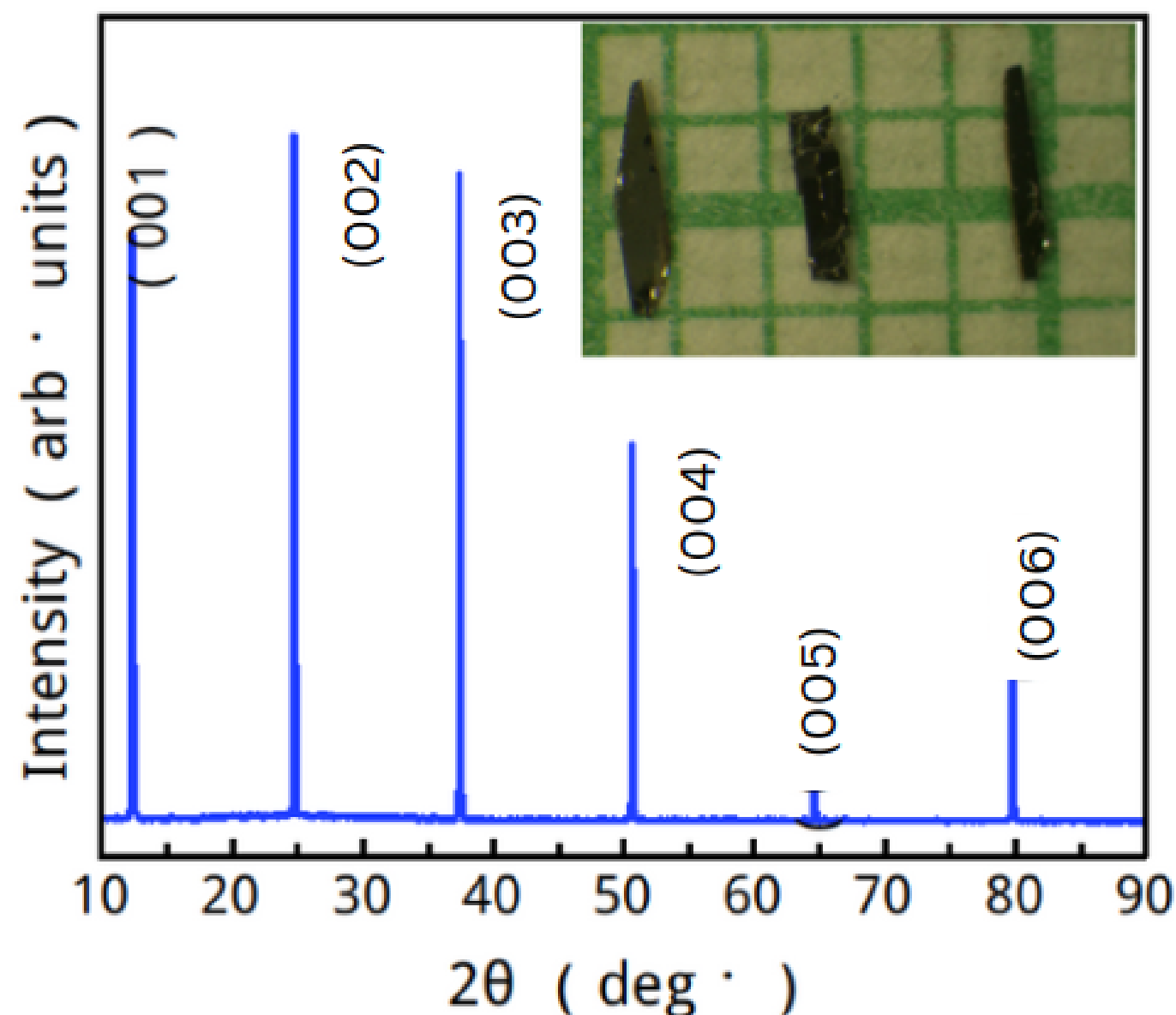


Figure: XRD pattern of the synthesized crystal, showing only (00l) peaks

Technique: X-Ray Diffraction (XRD) on the flat, cleaved surface of a single crystal.

Analysis & Conclusions:

- **High Crystallinity:** The peaks are sharp and well-defined, not broad, indicating a high-quality crystal.
- **Preferred orientation along the c-axis:** The pattern is dominated by a single family of peaks: (00l) (l=1, 2, 3, 4, 5, 6).

Characterization III: R(T) (Electronic Properties)

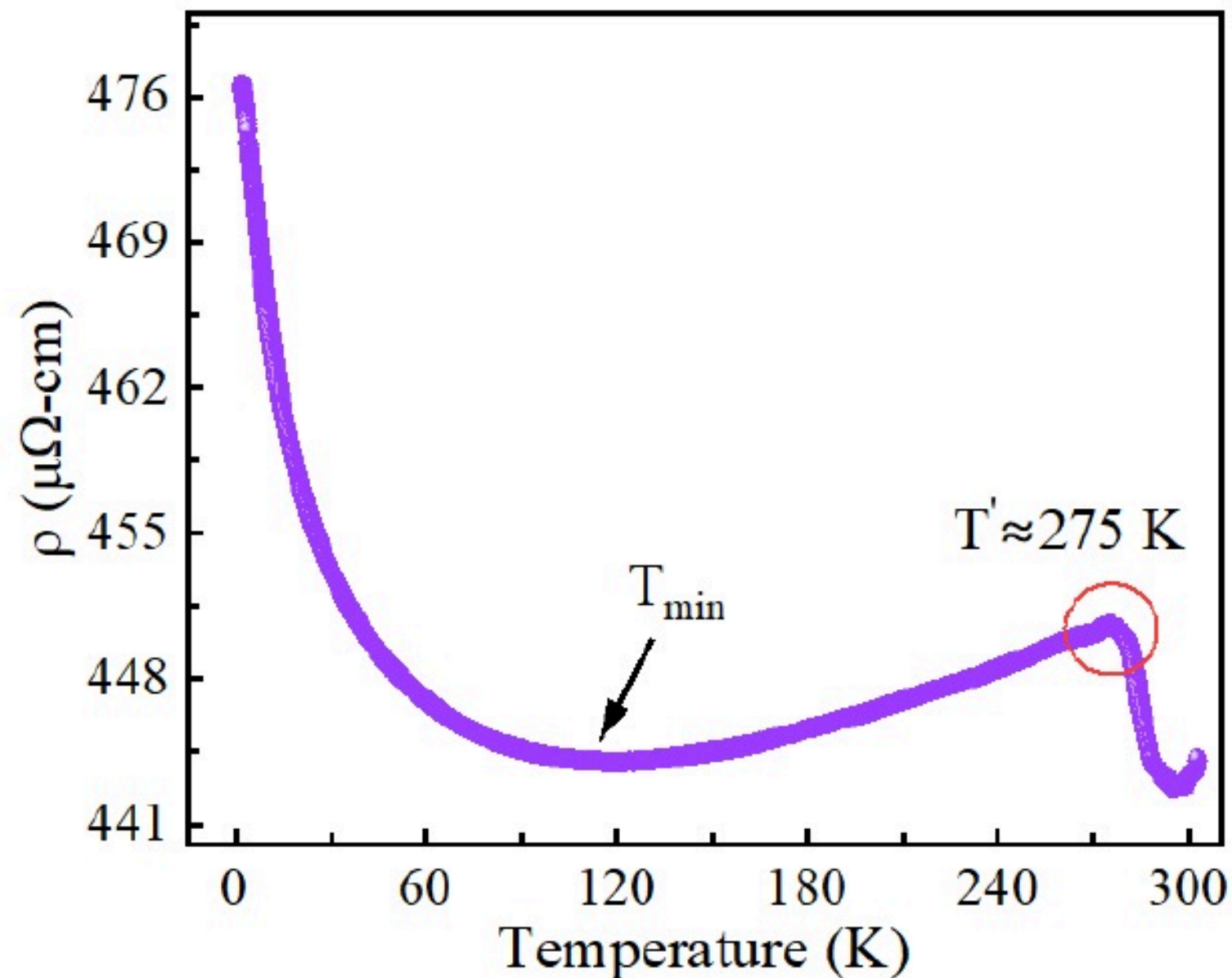
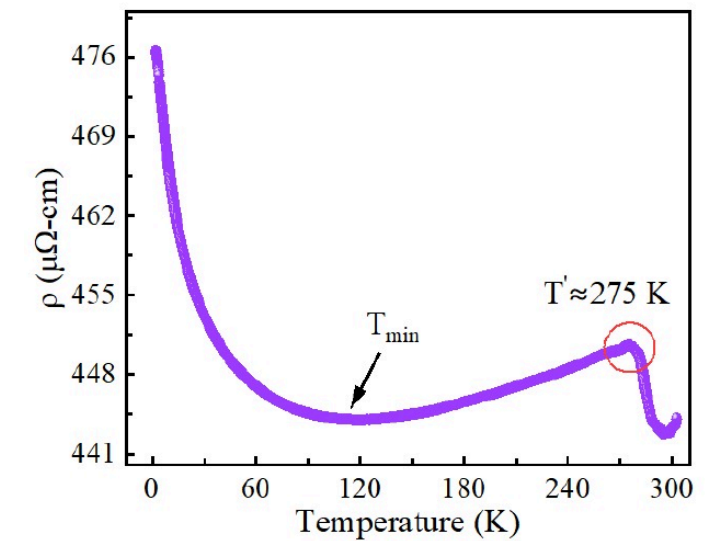
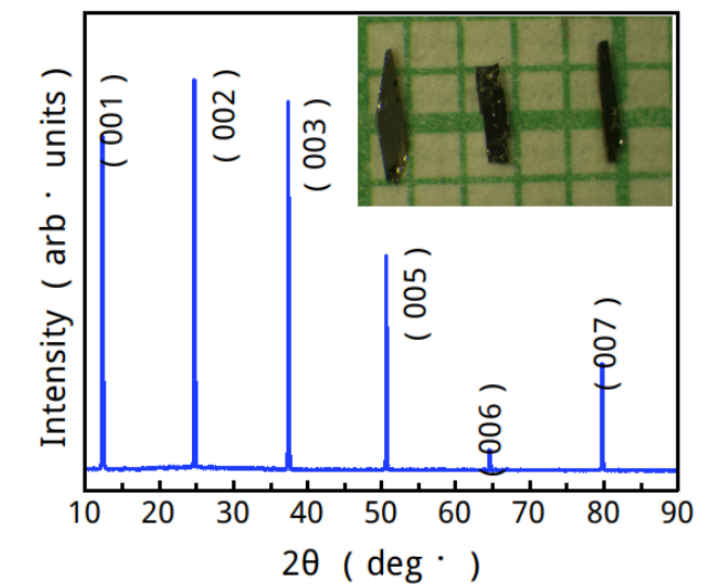
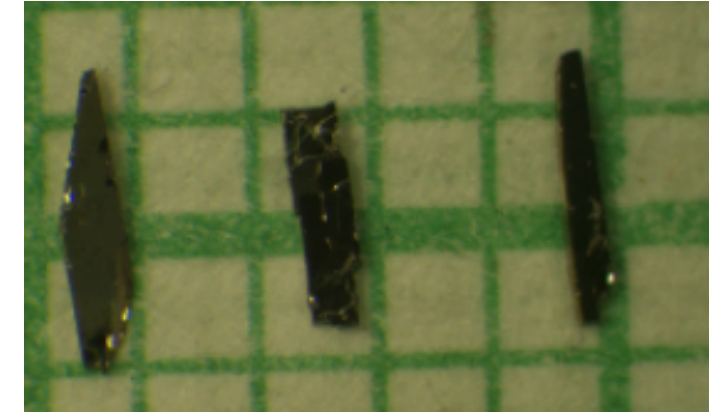


Figure: Resistivity v/s Temperature

- The material is metallic, with a low room-temperature resistivity. The behavior is highly anomalous.
- **Anomaly 1 (High T):** A sharp, steep drop in resistivity is observed at $T' \sim 275$ K. Indicator of a phase transition.
- **Anomaly 2 (Low T):** Below 111.5 K, resistivity increases as the temperature drops.
- It was observed that the heating and cooling curves are identical, implying that there is no thermal hysteresis.

Conclusion

1. **Synthesis:** Successfully synthesized high-quality, cleavable single crystals
2. **Composition (EDX):** The grown crystals are non-stoichiometric, with an average composition of Ni: 1 , Nb: 1.05 , Te: 2.45 . This is a Te-rich phase of the 1:1:2 (NbNiTe_2) structure.
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